

RNA-Seq Gene Expression for Cancer-Type Prediction

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Goal	Dataset Scale	Best Result
Cancer-type prediction Use gene-expression patterns to classify cancer types and cluster low-confidence samples.	20,531 features Each patient record contains 20,531 gene-expression values before dimensionality reduction.	92.5% cluster accuracy Top hierarchical settings (ward / average) outperformed most alternatives, while K-Means reached 91.6%.

Overview

This project investigated cancer type prediction using RNA-Seq gene expression data, with a particular focus on identifying samples that could not be confidently assigned to known classes. The approach combined *SVM classification* with *K-Means* and *Hierarchical Clustering*, allowing low-confidence samples to be further analysed rather than being forced into known categories. The workflow also included *KNN imputation*, *PCA-based dimensionality reduction*, and class balancing with *KMeans-SMOTE* and *ADASYN*. Results showed that the proposed hybrid approach improved class separability, with the best clustering configuration achieving **92.5%** cluster accuracy.

Methodological Pipeline

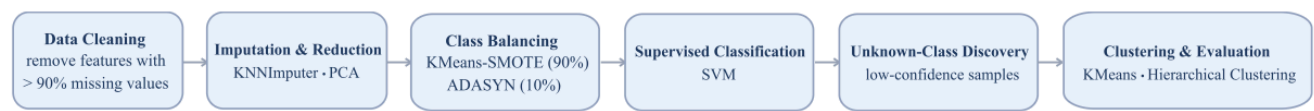


Figure 1. Proposed analytical pipeline for cancer type prediction: preprocessing and rebalancing were followed by SVM classification, while low-confidence samples (confidence < 0.95) were further analyzed using clustering.

- 1. Data cleaning:** The dataset originally contained **20,531 gene-expression features**, with a missing rate of **14.1%**. By removing features with more than **90% missing values**, the missing rate was reduced to **0.4%**, while feature dimensionality decreased by **22.6%**.
- 2. Imputation & Dimensionality reduction:**
The remaining missing values were imputed using *KNNImputer*. To address the curse of dimensionality, *PCA* was then applied to reduce the feature space to **292 principal components** while retaining **95%** of the total variance, yielding an overall dimensionality reduction of approximately **98.6%** relative to the original feature space.
- 3. Class balancing (Figure 2):**
Because the dataset was highly imbalanced across cancer types, a hybrid oversampling strategy was adopted using *KMeans-SMOTE (90%)* and *ADASYN (10%)*. This combination improved class balance while also reinforcing representation near difficult decision boundaries.

- 4. SVM-based classification:** *SVM* was selected as the primary classifier due to its strong performance on high-dimensional data. In internal validation on known classes, the model demonstrated very strong classification performance, and a confidence threshold of **0.95** was applied to identify uncertain predictions.
- 5. Unknown class discovery:** Samples with confidence scores below **0.95** were routed to unsupervised clustering for further analysis instead of being directly assigned to known categories. This design allowed the system to examine potentially unseen or ambiguous cancer classes outside the training set more cautiously.
- 6. Clustering and Evaluation:** Two clustering methods were compared: *K-Means* and *Hierarchical Clustering*. The results showed that *K-Means reached 91.6% accuracy*, while *Hierarchical Clustering achieved up to 92.5%*, although performance varied significantly across linkage strategies.

Post-PCA Class Distribution After Rebalancing



Figure 2. Post-PCA class distribution after hybrid rebalancing with *KMeans_SMOTE* (90%) + *ADASYN* (10%).

PCA and rebalancing improved class separability.

Figure 2 shows clearer class structure after dimensionality reduction and hybrid oversampling, indicating that the preprocessing pipeline enhanced the quality of the feature space.

Hybrid rebalancing strengthened performance.

The combination of *KMeans-SMOTE* (90%) and *ADASYN* (10%) improved class balance while preserving harder-to-learn samples near decision boundaries.

Cluster selection was not based on a single metric.

Both *Silhouette Analysis* and the *Elbow Method* were used to determine cluster counts before comparing clustering results across settings.

Clustering Performance by Method and Cluster Selection Criteria

Method	Clusters (Silhouette)	Accuracy	Clusters (Elbow)	Accuracy
K-Means	2	91.6%	2	91.6%
Hierarchical - single	2	53.3%	9	53.3%
Hierarchical - complete	2	92.2%	2	92.2%
Hierarchical - average	10	90.4%	6	92.5%
Hierarchical - ward	2	92.5%	2	92.5%

Table 1. Most hierarchical-linkage settings exceeded 90% accuracy, whereas single-linkage performed substantially worse. For single linkage, varying the selected number of clusters did not improve clustering accuracy. K-Means achieved 91.6% accuracy.

Relevance to Intelligent Robotics

Technical Insight	Robotics Relevance
Clustering performance varied substantially by linkage strategy: hierarchical clustering achieved up to 92.5% accuracy , whereas single linkage reached only 53.3% .	This comparison demonstrated the importance of algorithm selection, representation quality, and evaluation under different modelling assumptions when developing intelligent systems that must respond reliably to complex and uncertain inputs.
Reliable prediction required a structured pipeline for processing high-dimensional, incomplete, and imbalanced data, including cleaning, KNN imputation, PCA, class balancing, and confidence-based sample analysis.	This experience developed my ability to build robust learning pipeline, manage uncertainty, and analyse low-confidence outputs systematically-skills applicable to intelligent robotic systems operating with complex and imperfect data.